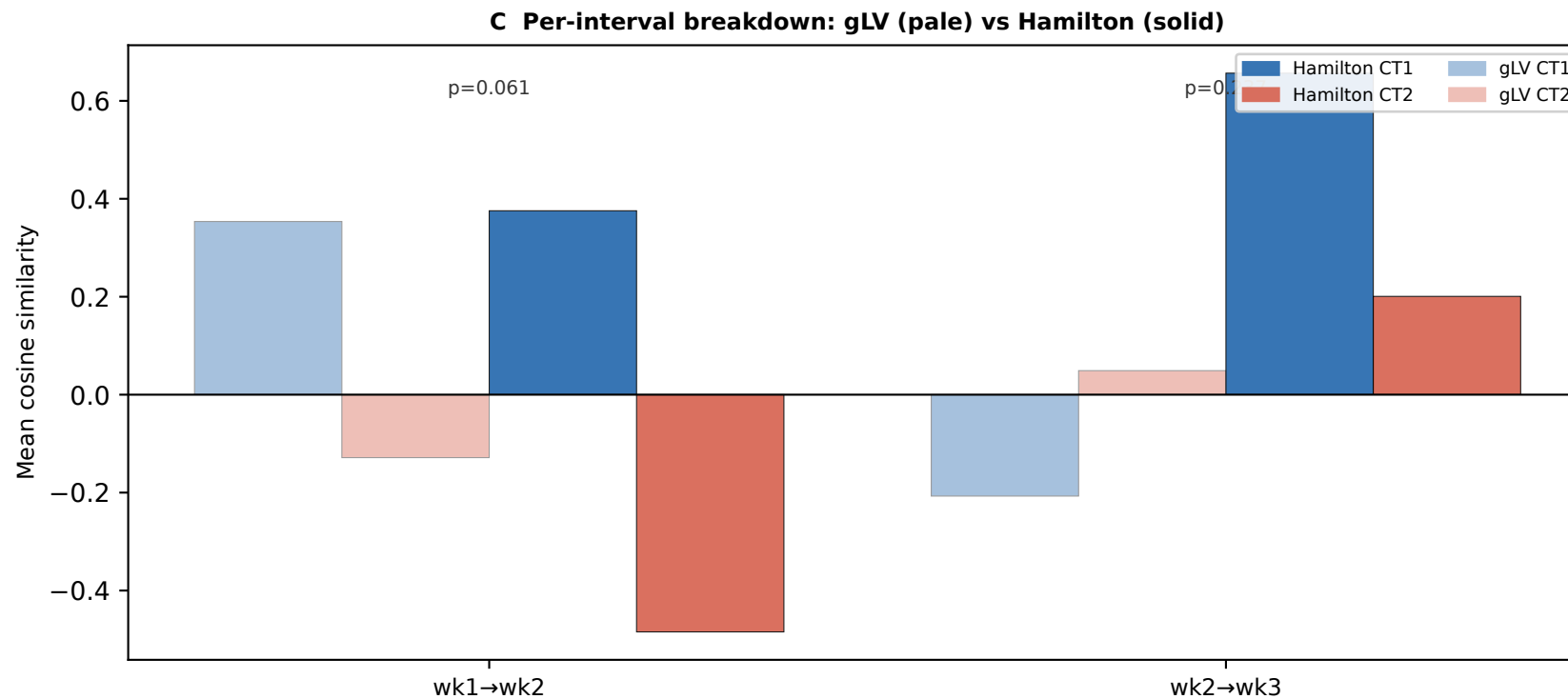
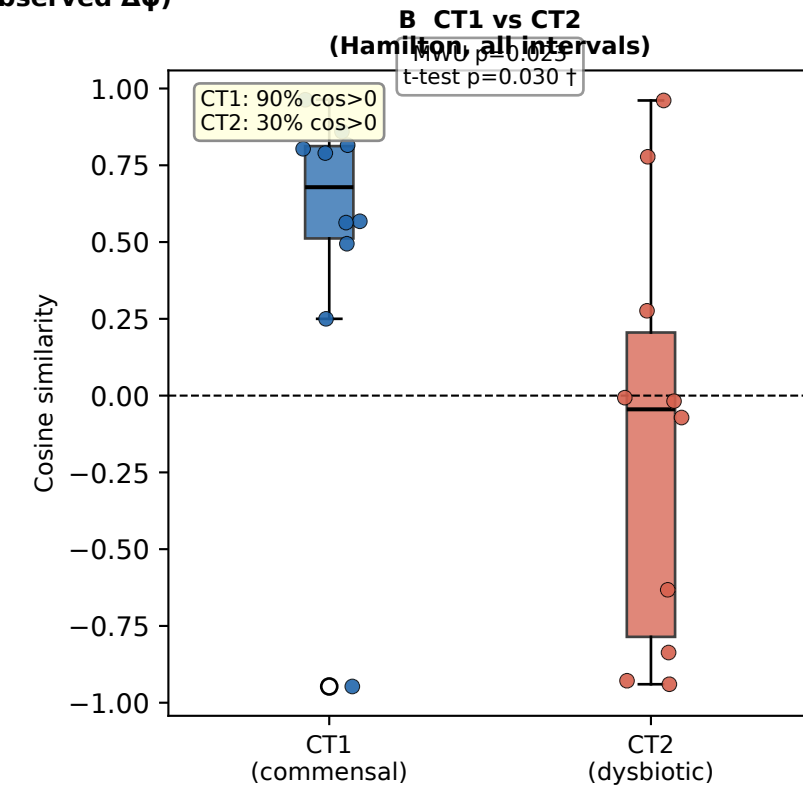
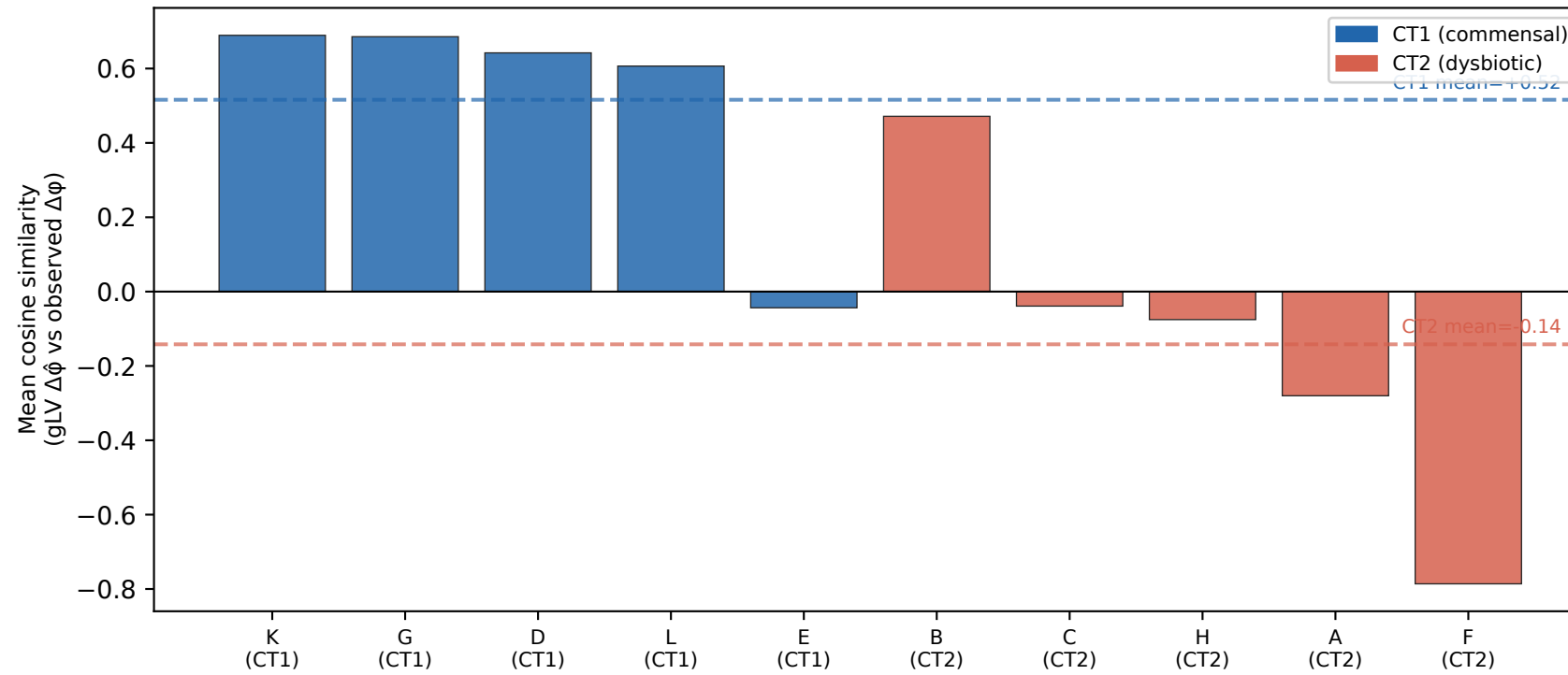


gLV velocity direction test: model is predictive in commensal basin, not in dysbiotic basin
 (Hamilton symmetric fit; per-patient b_i ; cosine similarity of predicted vs observed $\Delta\phi$)
A Hamilton fit: velocity direction accuracy per patient (mean over all intervals)
 $\cos > 0$ = model predicts correct direction of community change



Interpretation

CT1 (commensal) patients:
 90% of transitions predicted
 in correct direction (Hamilton)
 → gLV captures stabilizing
 dynamics near commensal basin

CT2 (dysbiotic) patients:
 Only 30% correct (Hamilton)
 → External forces dominate:
 immune response, treatment,
 dietary shifts
 → Ecological self-dynamics
 alone insufficient

Implication for model:
 AGORA metabolic priors needed
 to constrain dysbiotic basin
 (sign prior motivation ✓)